

Chapter 24

Radiation

24.1 Fields from non-zero acceleration

The acceleration piece of the Liénard–Wiechert fields is

$$\vec{E}_a = \frac{2q}{R c^2} \frac{\vec{m} \times \left(\left(\vec{m} - \frac{\vec{v}}{c} \right) \times \vec{a} \right)}{\left(1 - \frac{\vec{v} \cdot \vec{m}}{c} \right)^3} \quad , \quad \vec{B}_a = \vec{m} \times \vec{E}_a \quad (24.1)$$

where $\vec{y}$, $\vec{v}$, $\vec{a}$ are all evaluated at the retarded time t'

$$t - t' = \frac{1}{c} |\vec{x} - \vec{y}(t')| \quad (24.2)$$

24.2 Acceleration radiation and the Larmor formula

The acceleration radiation looks like electromagnetic waves:

$$\vec{m} \cdot \vec{E}_a = 0 \Rightarrow E_a = B_a \quad (24.3)$$

The power per solid angle is then

$$\frac{dP}{d\Omega} = r^2 \left(\hat{n} \cdot \vec{S} \right) = E_a^2 R^2 c \quad (24.4)$$

which is finite as $r \rightarrow \infty$. For $v \ll c$, this leads to the Larmor formula

$$P = \frac{q^2 a^2}{6\pi c^3} \quad (24.5)$$

with a evaluated at $t' = t - R/c$.

24.3 Radiation from a general source

For a general j^ν , one gets

$$A^\nu = \frac{1}{c} \int \frac{1}{|\vec{x} - \vec{y}|} j^\nu \left(t - \frac{|\vec{x} - \vec{y}|}{c}, \vec{y} \right) d^3y \quad (24.6)$$

Far from the source, we expand

$$|\vec{x} - \vec{y}| \rightarrow r - \hat{r} \cdot \vec{y} + \mathcal{O}\left(\frac{y^2}{r}\right) \quad , \quad \frac{1}{|\vec{x} - \vec{y}|} \rightarrow \frac{1}{r} + \mathcal{O}\left(\frac{y}{r^2}\right) \quad (24.7)$$

which gives

$$A^\nu = \frac{1}{r c} \int j^\nu (t - r + \hat{r} \cdot \vec{y}, \vec{y}) d^3y \quad (24.8)$$

24.4 Monochromatic sources

For a monochromatic source

$$j^\nu(t, \vec{x}) = e^{i\omega t} j^\nu(\omega, \vec{x}) + \text{c.c.} \quad (24.9)$$

the potential becomes

$$A^\nu(x) = e^{ikx} \varepsilon^\nu + \text{c.c.} \quad , \quad \varepsilon^\nu = \frac{1}{r} \int e^{i\omega \hat{r} \cdot \vec{y}} j^\nu(\omega, \vec{y}) d^3y \quad (24.10)$$

24.5 The linear antenna

For a linear antenna, take the current

$$\vec{J}(t, \vec{y}) = \delta(y^1) \delta(y^2) \frac{I_0}{\sin\left(\frac{\omega L}{2}\right)} \sin\left(\omega \left(\frac{L}{2} - |y^3|\right)\right) \cos(\omega t) \hat{z} \quad (24.11)$$

with $\vec{J} = 0$ for $|y^3| > L/2$, as depicted in the Figure. This gives

$$\vec{A} = \frac{I_0}{\sin\left(\frac{\omega L}{2}\right)} \frac{\cos \omega(t - r)}{2\pi r \omega \sin^2 \theta} \left[\cos\left(\frac{\omega L}{2} \cos \theta\right) - \cos\left(\frac{\omega L}{2}\right) \right] \hat{z} \quad (24.12)$$

so that the time-averaged power per solid angle is

$$\frac{d\bar{P}}{d\Omega} = \frac{I_0^2}{8\pi^2} \frac{\left(\cos\left(\frac{\omega L}{2} \cos \theta\right) - \cos\left(\frac{\omega L}{2}\right)\right)^2}{\sin^2 \frac{\omega L}{2} \sin^2 \theta} \quad (24.13)$$

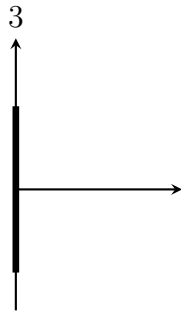


Figure 24.1: A linear antenna of length L along the z -axis.

This distribution vanishes at $\theta = 0$; the radiation is linearly polarized in the $\hat{z}$ - $\hat{r}$ plane; and it is maximal at $\theta = \pi/2$ if $L/\lambda \neq 2 \times \text{integer}$ (plots of the angular distribution were shown in lecture).